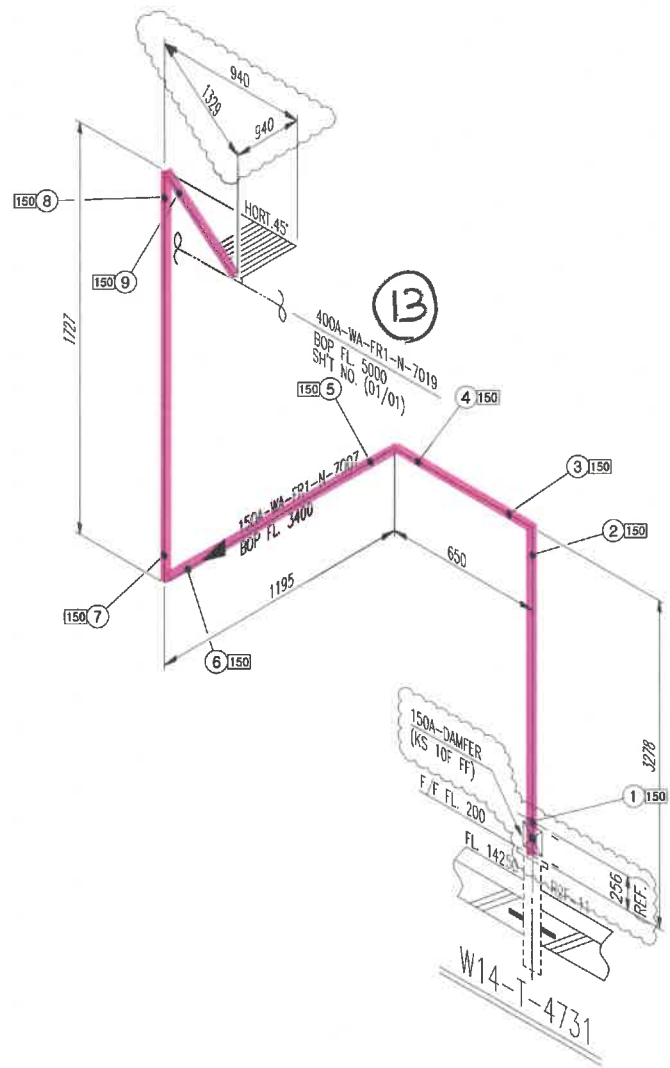


DRAWING LIST FOR TEST PACKAGE**WA-FR1-631****PROJECT NO.: 22328D****PROJECT TITLE : M15X Ph2 Project**

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	631	R2FA	150A-WA-FR1-N-7007-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
2	631	R2FA	200A-WA-FR1-N-7008-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
3	631	R2FA	200A-WA-FR1-N-7009-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
4	631	R2FA	150A-WA-FR1-N-7010-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
5	631	R2FA	150A-WA-FR1-N-7011-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
6	631	R2FA	150A-WA-FR1-N-7012-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
7	631	R2FA	150A-WA-FR1-N-7013-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
8	631	R2FA	150A-WA-FR1-N-7014-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
9	631	R2FA	150A-WA-FR1-N-7015-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
10	631	R2FA	200A-WA-FR1-N-7016-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 506	2A/1A	
11	631	R2FA	150A-WA-FR1-N-7017-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 506	2A/1A	
12	631	R2FA	150A-WA-FR1-N-7018-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 506	2A/1A	
13	631	R2FA	400A-WA-FR1-N-7019-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
14	631	2FA	400A-WA-FR1-N-7019-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
15	631	3FA	400A-WA-FR1-N-7019-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	



<NOTE>
1. TRAY & ANGLE CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * DOWNED : RC * ECG SWAGE : RES/RES1 * CON SWAGE : RCS/RCS1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET (HDPE/PVC) : SC * SADDLE (HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BOND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT (STEEL) : FLX * FLEXIBLE JOINT (RUBBER) : FLXR * FLEXIBLE JOINT (STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER (Y-TYPE) : YSTR * STRAINER (Z-TYPE) : ZSTR * STRAINER (BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020015-A	REF. P & ID	CJ-M15W14-PR-A20-000505	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60										25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
			OPERATING	4	AMB														

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	150A	6.2M	PPE FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	150A	4EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	150A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
G	150A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM THK
B	150A	2SET	MACHINE BOLT & NUTS :SIS 304 / W WASHERS

CLIENT	SK hynix	PROJECT NAME	M15X Ph4 Project
CONTRACTOR	SK ecoplant	JOB NO.	22330D
COMMODITY	WA	SH'T NO.	01 / 01
LINE NO.	150A-WA-FR1-N-7007	REV.	0



* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* COLLARED: HC	* ECC RED: RE	* CON SWAGE: RCS/RCS1	* ECC SWAGE: RES/RES1	* NIPPLE: NP/NP1/MP2	* CAP: CAP/CAP1	* FLUG: FLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET (HPIPE/PIPE): SC	* SADDLE (HPIPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VAL: VG/VG1/VG2	* BALL VAL: VB/VB1/VB2/VB3/VB4	* BUTTERFLY VAL: VF/VF1	* CHECK VAL: VC/VC1/VC2/VCS/VCS1/VCS2/VCS3	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHOWER: SSH		
* GLOBE VAL: VV/VV1/VV2	* DIAPHRAGM VAL: VD/VD1	* DAMPER: VP	* QUICK COUPLINGS: SC	* FLEXIBLE JOINT/STEEL: FLXJ	* FLEXIBLE JOINT/RUBBER: FLXJR	* ISOLATION JOINT: IJ	* FLEXIBLE JOINT/STEEL PIPE LINED: FLXJL	* FLEXIBLE HOSE: FLXH	* STEAM TRAP: ST	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT & NUT: BL/N1			
* STRAINER (T-YPE): TSTR	* STRAINER (H-TYPE): TSTR	* QUICK COUPLINGS: SC	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* EXHAUST KITT (KUBER): K	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTD: LOTD	* SIGHT GLASS: SG						

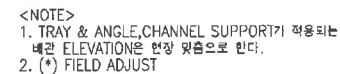
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA							M15X Ph4 Project		
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020014-A							COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000505							WA	22330D		
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
									25.01.24	APPROVED FOR CONSTRUCTION	황태훈	안남호	박동영	200A-WA-FR1-N-7008	01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단의 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유전자오류 불법 사용으로 피해가 발생할 경우 유전자오류에 민/형사상 책임을 요구할 수 있음.



* PIPE: P/P1	* 90° ELBOW: F90	* 45° ELBOW: F45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* COUPLER: RC	* ECC.RED: RE	* CON.SWAGE: RCS/RCS1	* ECC.SWAGE: RES/RES1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET HUBE: PHS/SC	* SOCKET HFPD: SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* RING FLANGE: FB/FB1	* GATE VA: VG/G1/G2	* BALL VA: VB/VG1/VB1/VB2/VB3	* BUTTERFLY VA: VF/VF1	* CHECK VA: VC/GA1/VG2/VG3/VG4/VG5/VG6/VG7					
* LOBE VA: VL/VL1/VL2	* DIAPHRAGM VA: VD/VD1	* DAMPER: VP	* QUICK COUPLER: FLEXIBLE JOINT/STEEL JOINT	* FLEXIBLE JOINT/ROTOR: FLX	* ISOLATION JOINT/ROTOR: FLX	* FLEXIBLE JOINT/STEEL PIPE LINED: FLX	* FLEXIBLE HOSE: FLXH	* FLEXIBLE HOSE: FLXH	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHUTTER: SSH				
* STRAINER(NP-TYPE): YSTR	* STRAINER(VP-TYPE): YSTR	* STRAINER(BLUE-TYPE): BSTR	* ANCHOR PLATE: STOP	* FLOX CABLE: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHALED: SH	* SPECIAL BOLT & NUT: BL/BL1			

경고 : 본 문서는 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단의 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* COMBED : RC	* ECC.RED : RE	* CON.SNAGE : RCS/RCS1	* ECC.SNAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLC	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1						
* STUB-IN : STI	* SOCKET/WOPE : SC	* SADDLE(WPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BRND FLANGE : FB/B1	* GATE VA : VG/G1/VG2	* BALL VA : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VV/VV1/VV2	* CHECK VA : VC/VC1/VCC/VCC1/VCC2/VCC21	* STRAIN INH : VI/VI1/VI2	* DAMPHRAG VA : VD/VD0	* DRAINER : VP	* QUICK COUPLING : QC	* FLEXIBLE JOINT(STEE) : FLXB	* ISOLATION JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL TYPE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHUTTER : SSH
* STRAIN INH : YSTR	* STRAIN INH-TYPE1 : YSTR1	* DAMPER(WOPE-TYPE) : YSTR2	* ANCHOR PLATE : STOP	* GRID SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTTO : LOT0	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHALED : SH	* SPECIAL BOLT & NUT : BL/BL1									

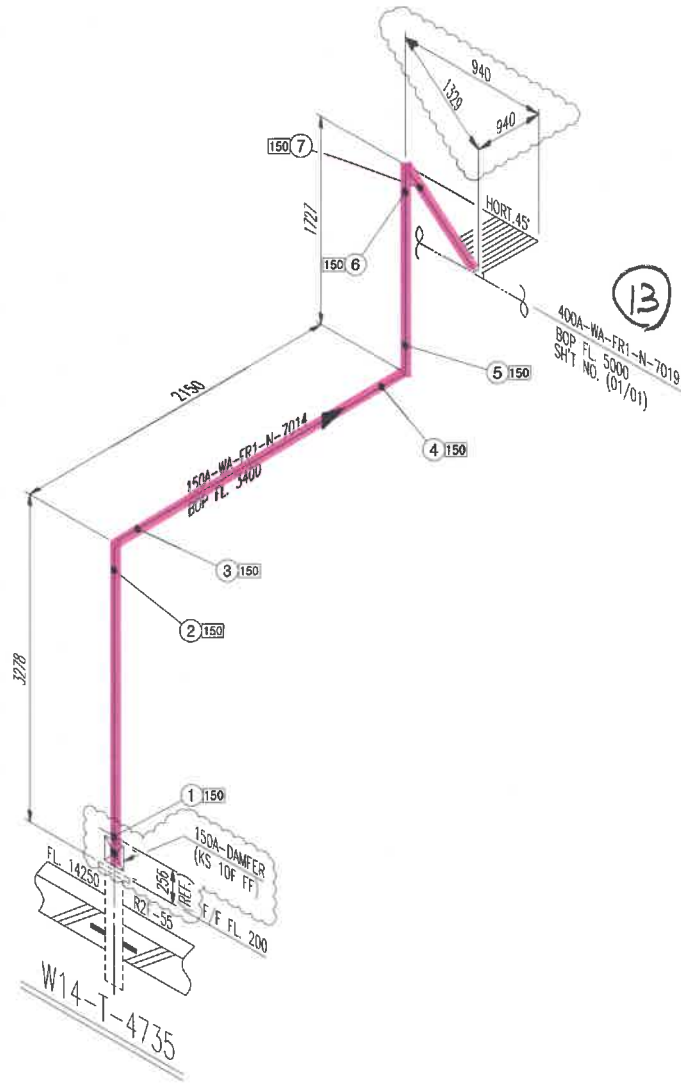
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	6.4	60	UNIT R2FA							M15X Ph4 Project		
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020014-A						COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000505	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
								25.01.24	APPROVED FOR CONSTRUCTION	황재중	안남후	박동영	150A-WA-FR1-N-7011	01 / 01	0

참고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단의 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.



0	25.01.24	APPROVED FOR CONSTRUCTION	황래준	안남후	박동
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.	150A-WA-FR1-N-7013	SH'T NO. 01 / 01	REV. 0



<NOTE>
1. TRAY & ANGLE CHANNEL SUPPORT기 적용되는
배관 ELEVATION은 현장 모습으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLUED : RC * ECC RED : RE * CON SWAGE : RCS/RCS1 * ECC SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PPVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHUTTER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020014-A	REF. P & ID	CJ-M15W14-PR-A20-000505	REV.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60										25.01.24	APPROVED FOR CONSTRUCTION	황재홍	안남후	박동영
			OPERATING	4	AMB														

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	150A	6.9M	PIPE, FRP(MANUFACTURER STD), PE/BUT-JOINT, 4.8T
E90	150A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	150	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
G	150A	2EA	GASKET, KS 10K, FF, EPDM, 3.DMM TH'K
B	150A	2SET	MACHINE BOLT & NUTS :SIS 304 / W WASHERS

CLIENT

SK hynix

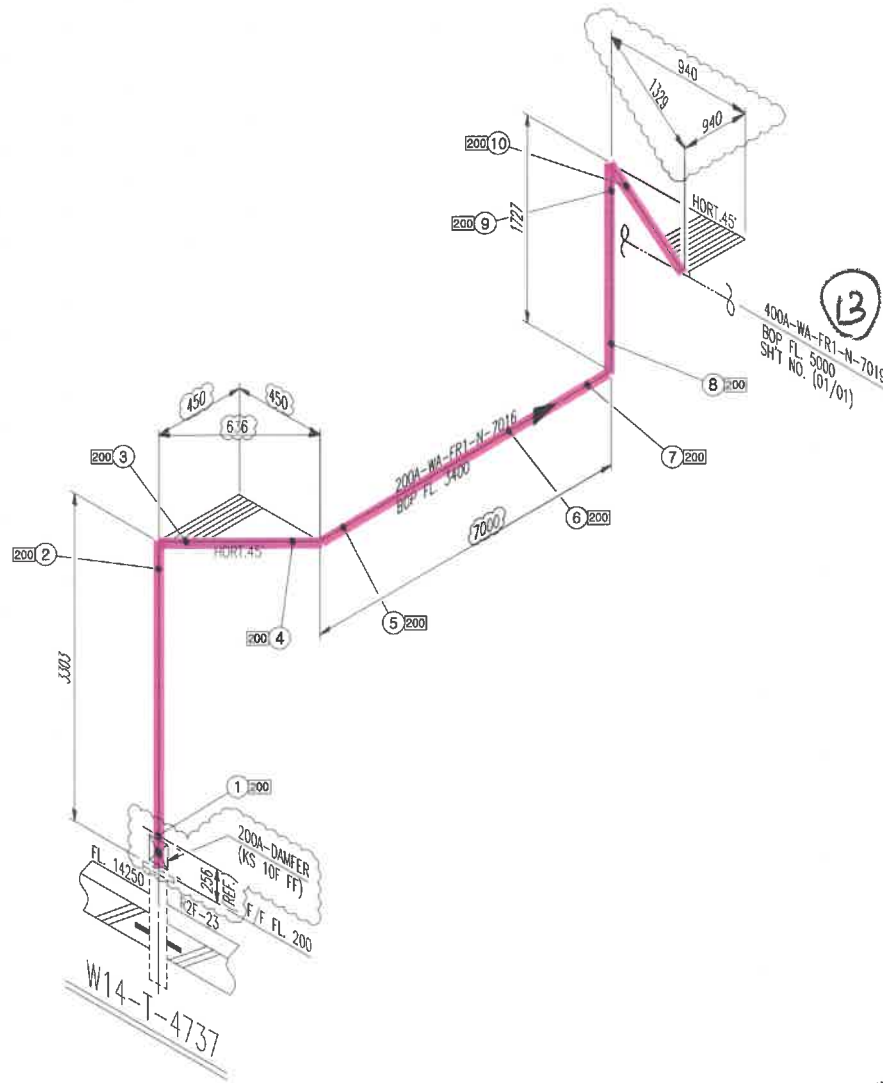
CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	WA	JOB NO.	22330D
LINE NO.	150A-WA-FR1-N-7014	SH'T NO.	01 / 01
		REV.	0



* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET (HPIPE/PVC) : SC * SADDLE (HPIPE) : SD * FLANGE : F/F1 * CASSETT : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/G1A/G2 * BALL VA. : VB/G1A/B1/B1B1A/B2/B2B2A/B2B2B2 * BUTTERFLY VA. : VF/VF1A * CHECK VA. : VC/G1A/AC2/AC3/AC4/AC5/AC6/AC7/AC8
* GLOBE VA. : VL/VL1/VL2 * (DIAPHRAGM VA. : VD/VD1) * DAMPER : VP * QUICK COUPLING : F * FLEXIBLE JOINT/STEEL : FLX/FLXB * FLEXIBLE JOINT/RUBBER : FLXR * FLEXIBLE JOINT/IRON PIPE UNED : FLXU * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST
* STRAIN/V-TYPE : VTY * STRAIN/TIE-TYPE : ISTR * STRAIN/REDUCT-TYPE : ISTR * ANCHOR PLATE : STB * FLOOR CLAMP : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/NL



CLIENT		 	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.		SH'T NO.	REV.
150A-WA-FR1-N-7015		01 / 01	0



<NOTE>
 1. TRAY & ANGLE CHANNEL SUPPORT기 적용되는
 배관 ELEVATION은 현장 맞춤으로 한다.
 2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * SUB-IN : STU * REINFORCING PAD : RP * CONVED : RC * ECC RED : RE * CON SWAGE : RCS/W/S1 * ECC SWAGE : RCS/H/S1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * SUB-IN : STU * SOCKET(HOPE/PVC) : SC * SADDLE(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
 * GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLX * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER(I-TYPE) : YSTR * STRAINER(L-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BRID. SCRTEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020015-A	REF. P & ID	CJ-M15W14-PR-A20-000505,000506	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60									25.01.24	APPROVED FOR CONSTRUCTION	황재홍	안남후	박동영
			OPERATING	4	AMB													

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	200A	11.9M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8I
E90	200A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8I
E45	200A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2I
F	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / CRWG-EPDM, LEVER
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS-STS 304 / W WASHERS

CLIENT

SK hynix

CONTRACTOR

SK ecoplant

COMMODITY

WA

LINE NO.

200A-WA-FR1-N-7016

PROJECT NAME

M15X Ph4 Project

JOB NO.

22330D

SH'T NO.

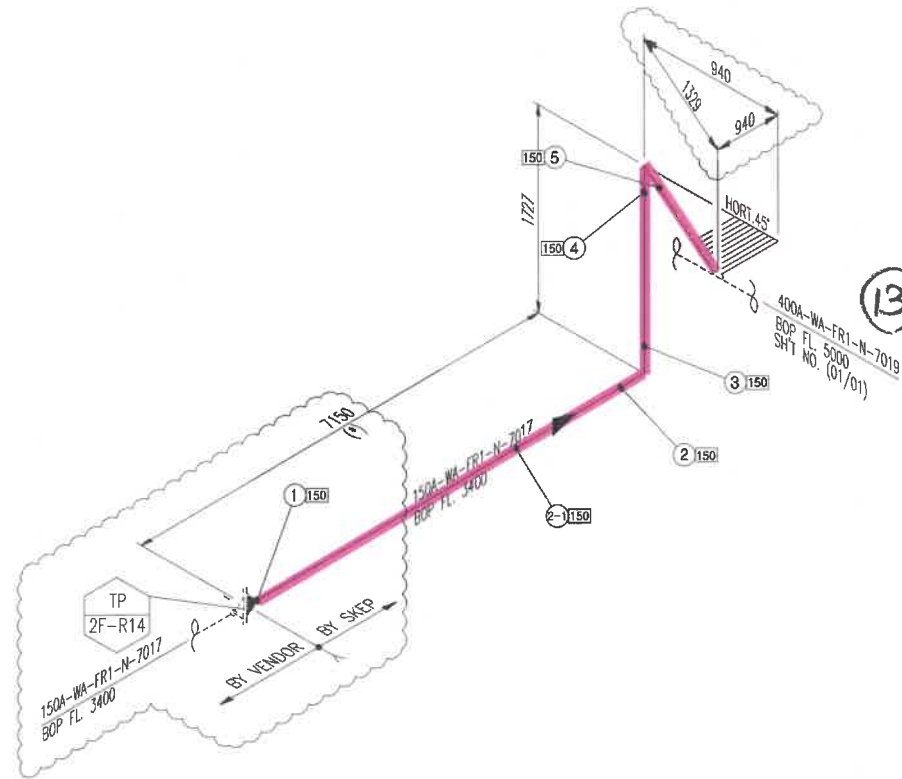
01 / 01

REV.

0



ITEM	SIZE	Q'TY	DESCRIPTION
		FR1	
P	150A	9.2M	PIPE, FRP(MANUFACTURER STD), PE/BIUT-JOINT, 4.8T
EBD	150A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTI-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTI-JOINT



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

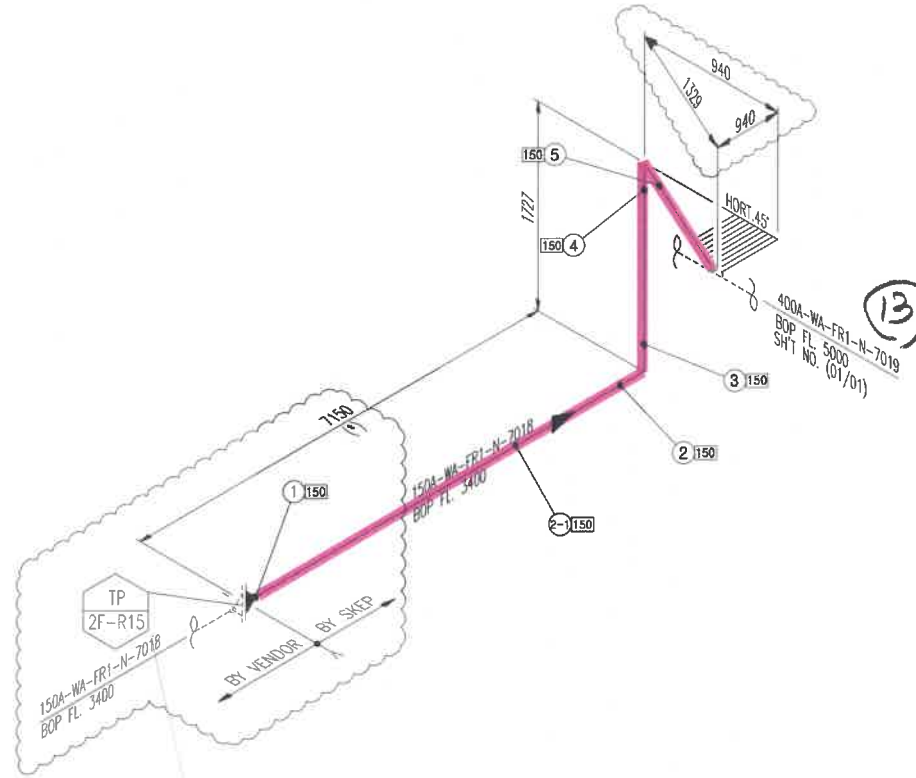
* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* COURED: RC	* ECC.RED: RE	* CON.SNAGE: RCS/RCS1	* ECC.SNAGE: RES/RES1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET(HDPE,PMC): SC	* SADD(HDPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VA: VG/N1/N2	* BALL VA: VB/N1/VB1/VB10/A2/B2/B20	* BUTTERFLY VA: VF/VF1	* CHECK VA: VC/N1/VC1/VC10/A3/A3A/NCS/M62/N62					
* GLOBE VA: V1/M1/M2	* DIAPHRAGM VA: VA/VA0	* DAMPER: VP	* JLUCK COUPLER: F	* FLEXIBLE JOINT(STEEL): FLX	* FLEXIBLE JOINT(RUBBER): FLXR	* ISOLATION JOINT: IK	* UNLOADING BOX: JB	* AIR TRAP: AT	* LOT0: LOT0	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT & NUT: BL/BL1		
* STRAINER(N-TYPE): YSTR	* STRAINER(H-TYPE): ISTR	* DAMPER(BLOCK-TYPE): BSIR	* ANCHOR PLATE: STOP	* FLOOR COVER: BS											

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA							M15X Ph4 Project	
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020014-A							COMMODITY	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000505,000506							WA	223300	
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	150A-WA-FR1-N-7017	SH'T NO. 01 / 01	REV. 0

※고지 : 본 두면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	150A	9.2M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	150A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT기 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : SIU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUS : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : SIU * SOCKET(HDPE/PMMA) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FTB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLVL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(W-Y-TYPE) : YSTR * STRAINER(V-TYPE) : VSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTD * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14				
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA				
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020014-A				
						REF. P & ID	CJ-M15W14-PR-A20-000505,000506				

REV
NO

25.01.24
DATE

APPROVED FOR CONSTRUCTION
DESCRIPTION

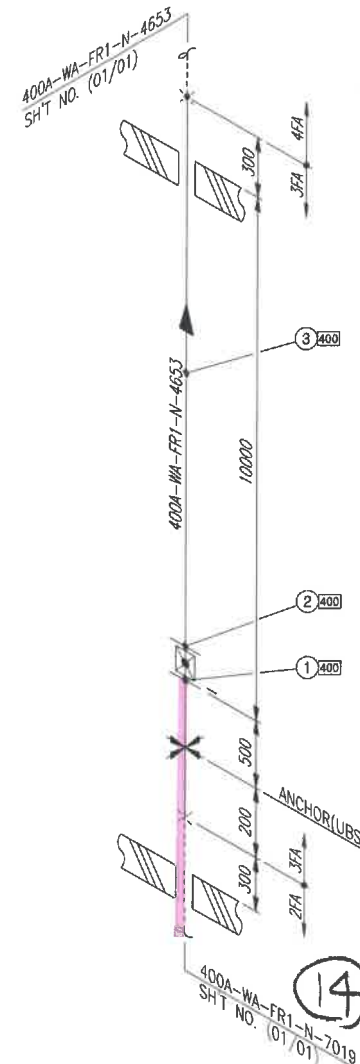
BY

CHK

APP'D

CLIENT

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
WA	223300
LINE NO.	SH'T NO.
150A-WA-FR1-N-7018	01 / 01
	REV.
	0



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED. TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON. RED. : RC * ECC. RED. : RE * CON. SWAGE : RCS/RCS1 * ECC. SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET (HDPE/PMC) : SC * SADDLE (HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LD * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT (STEEL) : FLXB * FLEXIBLE JOINT (RUBBER) : FLXR * FLEXIBLE JOINT (STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER (I-TYPE) : YSTR * STRAINER (T-TYPE) : TSTR * STRAINER (BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTD : LOTD * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								
	THICK.	-	DESIGN	1.0	AMB	UNIT	3FA								
			OPERATING	ATM	AMB	REF. PLAN DWG.									
						CJ-M15W14-PI-A30-030007-A									
						REF. P & ID									
						CJ-M15W14-PR-A20-000168									

REV. NO.

25.01.24

APPROVED FOR CONSTRUCTION

함재범

안남우

박동영

DATE

DESCRIPTION

BY

CHK

APP'D

CONTRACTOR

SK ecoplant

COMMODITY

WA

LINE NO.

400A-WA-FR1-N-4653

PROJECT NAME

M15X Ph4 Project

JOB NO.

223300

SH'T NO.

01 / 01

REV.

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경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해를 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

PUNCH LIST

Project No : 22330D

Project Title : M15X Ph3 Project

Package No :

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning